

SOLIDWORKS API (.NET VBA C#) — Reference

(75/337): IDisplayData

SOLIDWORKS API Documentation (converted from CHM)

IDisplayData (Type)

IDisplayData Interface

[See Also](#) [Members](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) : IDisplayData Interface

Allows access to display information for certain items, including reference planes and reference axes shown in a drawing view.

NOTE: Click the **Members** link, located near the top of the topic, to see this interface's methods and properties.

.NET Syntax

Visual Basic (Declaration)

```
Public Interface IDisplayData
```

Visual Basic (Usage)



```
Dim instance As IDisplayData
```

C#

```
public interface IDisplayData
```

C++/CLI

```
public interface class IDisplayData
```

Visual Basic for Applications (VBA) Syntax

See [DisplayData](#).

Example

[Get Centerline Annotation Information \(VBA\)](#)
[Get the Number of Lines in Flat-pattern Drawing View's Boundary-box Sketch \(C#\)](#)
[Get the Number of Lines in Flat-pattern Drawing View's Boundary-box Sketch \(VB.NET\)](#)
[Get the Number of Lines in Flat-pattern Drawing View's Boundary-box Sketch \(VBA\)](#)

Accessors

[IAnnotation::GetDisplayData](#) and [IAnnotation::IGetDisplayData](#)
[IBomTable::GetDisplayData](#) and [IBomTable::IGetDisplayData](#)
[ICThread::GetDisplayData](#) and [ICThread::IGetDisplayData](#)
[IDisplayDimension::GetDisplayData](#) and [IDisplayDimension::IGetDisplayData](#)
[IMultiJogLeader::GetDisplayData](#) and [IMultiJogLeader::IGetDisplayData](#)
[IView::GetDisplayData3](#) and [IView::IGetDisplayData3](#)
[IView::GetSMBoundingBoxDisplayData](#) and [IView::IGetSMBoundingBoxDisplayData](#)

Access Diagram

[DisplayData](#)

See Also

[IDisplayData Members](#)
[SolidWorks.Interop.sldworks Namespace](#)

IDisplayData Interface Members

[See Also](#) [Methods](#)



[SolidWorks.Interop.sldworks Namespace](#) : IDisplayData Interface

The following tables list the members exposed by [IDisplayData](#).

Public Methods

Name	Description
 GetArcAtIndex	Obsolete. Superseded by IDisplayData.GetArcAtIndex2 .
 GetArcAtIndex2	Gets information for the specified arc.
 GetArcCount	Gets the number of arc items.
 GetArrowHeadAtIndex	Obsolete. Superseded by IDisplayData.GetArrowheadAtIndex2 .
 GetArrowHeadAtIndex2	Gets information on the specified arrow head.
 GetArrowHeadCount	Gets the number of arrow heads.
 GetEllipseAtIndex	Obsolete. Superseded by IDisplayData.GetEllipseAtIndex2 .
 GetEllipseAtIndex2	Gets information for the specified ellipse.
 GetEllipseCount	Gets the number of line ellipses.
 GetLineAtIndex	Obsolete. Superseded by IDisplayData.GetLineAtIndex2 .
 GetLineAtIndex2	Obsolete. Superseded by IDisplayData.GetLineAtIndex3 .
 GetLineAtIndex3	Gets information for the specified line.
 GetLineCount	Gets the number of lines.
 GetParabolaAtIndex	Gets information for the specified parabola.
 GetParabolaCount	Gets the number of parabolas.
 GetPointAtIndex	Gets information about the specified point.
 GetPointCount	Gets the number of points.
 GetPolygonAtIndex	Gets information for the specified polygon.
 GetPolygonCount	Gets the number of polygons.
 GetPolygonSizeAtIndex	Gets the number of array elements returned by IDisplayData.GetPolygonAtIndex and IDisplayData.IGetPolygonAtIndex .
 GetPolylineAtIndex	Obsolete. Superseded by IDisplayData.GetPolylineAtIndex2 .
 GetPolylineAtIndex2	Gets information for the specified polyline.
 GetPolyLineCount	Gets the number of polylines.
 GetPolylineSizeAtIndex	Obsolete. Superseded by IDisplayData.GetPolylineSizeAtIndex2 .
 GetPolylineSizeAtIndex2	Gets the number of array elements returned by IDisplayData.GetPolylineAtIndex2 and IDisplayData.IGetPolylineAtIndex2 .
 GetTextAngleAtIndex	Gets the text angle for the specified piece of text.
 GetTextAtIndex	Gets the specified text string.
 GetTextCount	Gets the number of text items.
 GetTextFontAtIndex	Gets the specified text font.
 GetTextHeightAtIndex	Gets the text height for the specified piece of text .
 GetTextInBoxHeightAtIndex	Gets the text-in-box height of the specified text and table cell in a table annotation.
 GetTextInBoxStyleAtIndex	Gets the text-in-box style of the specified text and table cell in a table annotation.
 GetTextInBoxWidthAtIndex	Gets the text-in-box width of the specified text and table cell in the table annotation.
 GetTextInvertAtIndex	Gets the specified text's invert flag.
 GetTextLineSpacingAtIndex	Gets the line spacing for this text.
 GetTextPlaneAtIndex	Gets the rotation matrix of the specified text relative to the X-Y plane of the model.
 GetTextPositionAtIndex	Gets the text item offset relative to the note text point.
 GetTextRefPositionAtIndex	Gets the specified text item reference position.
 GetTriangleAtIndex	Gets the triangle at the specified index.

	GetTriangleCount	Gets the number of triangles.
	IGetArcAtIndex	Obsolete. Superseded by IDisplayData::IGetArcAtIndex2 .
	IGetArcAtIndex2	Gets information for the specified arc.
	IGetArrowHeadAtIndex	Gets information on the specified arrow head.
	IGetEllipseAtIndex	Obsolete. Superseded by IDisplayData::IGetEllipseAtIndex2 .
	IGetEllipseAtIndex2	Gets information for the specified ellipse.
	IGetLineAtIndex	Obsolete. Superseded by IDisplayData::IGetLineAtIndex2 .
	IGetLineAtIndex2	Obsolete. Superseded by IDisplayData::IGetLineAtIndex3 .
	IGetLineAtIndex3	Gets information for the specified line.
	IGetParabolaAtIndex	Gets the information for the specified parabola.
	IGetPointAtIndex	Gets information about the specified point.
	IGetPolygonAtIndex	Gets information for the specified polygon.
	IGetPolylineAtIndex	Obsolete. Superseded by IDisplayData::IGetPolylineAtIndex2 .
	IGetPolylineAtIndex2	Gets information for the specified polyline.
	IGetTextPositionAtIndex	Gets the text item offset relative to the note text point.
	IGetTriangleAtIndex	Gets the triangle at the specified index.

[Top](#)

See Also

[IDisplayData Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)

SOLIDWORKS API Help

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IDisplayData Interface Methods

[See Also](#)

	SolidWorks.Interop.sldworks Namespace : IDisplayData Interface
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For a list of all members of this type, see [IDisplayData members](#).

Public Methods

Name	Description
	GetArcAtIndex Obsolete. Superseded by IDisplayData::GetArcAtIndex2 .
	GetArcAtIndex2 Gets information for the specified arc.
	GetArcCount Gets the number of arc items.
	GetArrowHeadAtIndex Obsolete. Superseded by IDisplayData::GetArrowheadAtIndex2 .
	GetArrowHeadAtIndex2 Gets information on the specified arrow head.
	GetArrowHeadCount Gets the number of arrow heads.
	GetEllipseAtIndex Obsolete. Superseded by IDisplayData::GetEllipseAtIndex2 .
	GetEllipseAtIndex2 Gets information for the specified ellipse.
	GetEllipseCount Gets the number of line ellipses.
	GetLineAtIndex Obsolete. Superseded by IDisplayData::GetLineAtIndex2 .
	GetLineAtIndex2 Obsolete. Superseded by IDisplayData::GetLineAtIndex3 .
	GetLineAtIndex3 Gets information for the specified line.
	GetLineCount Gets the number of lines.
	GetParabolaAtIndex Gets information for the specified parabola.
	GetParabolaCount Gets the number of parabolas.
	GetPointAtIndex Gets information about the specified point.

		
	GetPointCount	Gets the number of points.
	GetPolygonAtIndex	Gets information for the specified polygon.
	GetPolygonCount	Gets the number of polygons.
	GetPolygonSizeAtIndex	Gets the number of array elements returned by IDisplayData::GetPolygonAtIndex and IDisplayData::!GetPolygonAtIndex .
	GetPolylineAtIndex	Obsolete. Superseded by IDisplayData::GetPolylineAtIndex2
	GetPolylineAtIndex2	Gets information for the specified polyline.
	GetPolyLineCount	Gets the number of polylines.
	GetPolylineSizeAtIndex	Obsolete. Superseded by IDisplayData::GetPolylineSizeAtIndex2 .
	GetPolylineSizeAtIndex2	Gets the number of array elements returned by IDisplayData::GetPolylineAtIndex2 and IDisplayData::!GetPolylineAtIndex2 .
	GetTextAngleAtIndex	Gets the text angle for the specified piece of text.
	GetTextAtIndex	Gets the specified text string.
	GetTextCount	Gets the number of text items.
	GetTextFontAtIndex	Gets the specified text font.
	GetTextHeightAtIndex	Gets the text height for the specified piece of text .
	GetTextInBoxHeightAtIndex	Gets the text-in-box height of the specified text and table cell in a table annotation.
	GetTextInBoxStyleAtIndex	Gets the text-in-box style of the specified text and table cell in a table annotation.
	GetTextInBoxWidthAtIndex	Gets the text-in-box width of the specified text and table cell in the table annotation.
	GetTextInvertAtIndex	Gets the specified text's invert flag.
	GetTextLineSpacingAtIndex	Gets the line spacing for this text.
	GetTextPlaneAtIndex	Gets the rotation matrix of the specified text relative to the X-Y plane of the model.
	GetTextPositionAtIndex	Gets the text item offset relative to the note text point.
	GetTextRefPositionAtIndex	Gets the specified text item reference position.
	GetTriangleAtIndex	Gets the triangle at the specified index.
	GetTriangleCount	Gets the number of triangles.
	!GetArcAtIndex	Obsolete. Superseded by IDisplayData::!GetArcAtIndex2
	!GetArcAtIndex2	Gets information for the specified arc.
	!GetArrowHeadAtIndex	Gets information on the specified arrow head.
	!GetEllipseAtIndex	Obsolete. Superseded by IDisplayData::!GetEllipseAtIndex2
	!GetEllipseAtIndex2	Gets information for the specified ellipse.
	!GetLineAtIndex	Obsolete. Superseded by IDisplayData::!GetLineAtIndex2 .
	!GetLineAtIndex2	Obsolete. Superseded by IDisplayData::!GetLineAtIndex3
	!GetLineAtIndex3	Gets information for the specified line.
	!GetParabolaAtIndex	Gets the information for the specified parabola.
	!GetPointAtIndex	Gets information about the specified point.
	!GetPolygonAtIndex	Gets information for the specified polygon.
	!GetPolylineAtIndex	Obsolete. Superseded by IDisplayData::!GetPolylineAtIndex2
	!GetPolylineAtIndex2	Gets information for the specified polyline.
	!GetTextPositionAtIndex	Gets the text item offset relative to the note text point.
	!GetTriangleAtIndex	Gets the triangle at the specified index.

[Top](#)

GetArcAtIndex Method (IDisplayData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IDisplayData Interface](#) : GetArcAtIndex Method (IDisplayData)

Obsolete. Superseded by [IDisplayData::GetArcAtIndex2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function GetArcAtIndex( _  
    ByVal Index As System.Integer _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IDisplayData  
Dim Index As System.Integer  
Dim value As System.Object  
  
value = instance.GetArcAtIndex(Index)
```

C#

```
System.object GetArcAtIndex(  
    System.int Index  
)
```

C++/CLI

```
System.Object^ GetArcAtIndex(  
& System.int Index  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Index

Visual Basic for Applications (VBA) Syntax

See [DisplayData::GetArcAtIndex](#).

See Also

[IDisplayData Interface](#)
[IDisplayData Members](#)

GetArcAtIndex2 Method (IDisplayData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IDisplayData Interface](#) : GetArcAtIndex2 Method (IDisplayData)

Gets information for the specified arc.

.NET Syntax

Visual Basic (Declaration)

```
Function GetArcAtIndex2( _  
    ByVal Index As System.Integer _  
) As System.Object
```

Visual Basic (Usage)

```
Dim instance As IDisplayData
Dim Index As System.Integer
Dim value As System.Object

value = instance.GetArcAtIndex2(Index)
```

C#

```
System.Object GetArcAtIndex2(
    System.int Index
)
```

C++/CLI

```
System.Object^ GetArcAtIndex2(
    & System.int Index
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Index
Index of the desired arc where the index begins at zero

Return Value

Array of doubles (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [DisplayData::GetArcAtIndex2](#).

Remarks

The return value is the following array of doubles :

[*color*, *lineType*, *Unused*, *Unused*, *startPt*[3], *endPt*[3], *centerPt*[3], *arcNormal*[3], *rotationDir*]

where:

<i>color</i>	COLORREF returned as an integer; return value can be 0 or -1 for default color
<i>LineType</i>	Line type as defined in swLineTypes_e
<i>startPt</i> [3]	XYZ arc start point
<i>endPt</i> [3]	XYZ arc end point
<i>centerPt</i> [3]	XYZ arc center point
<i>arcNormal</i> [3]	XYZ arc normal vector
<i>rotationDir</i>	Boolean returned as a double that represents the rotation direction, where CCW = True and CW = false

See Also

[IDisplayData Interface](#)
[IDisplayData Members](#)
[IDisplayData::GetLineAtIndex2 Method](#)
[IDisplayData::GetArcCount Method](#)

GetArcCount Method (IDisplayData)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IDisplayData Interface](#) : GetArcCount Method (IDisplayData)

Gets the number of arc items.

.NET Syntax

Visual Basic (Declaration)

```
Function GetArcCount() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IDisplayData  
Dim value As System.Integer  
  
value = instance.GetArcCount()
```

C#

```
System.int GetArcCount()
```

C++/CLI

```
System.int GetArcCount();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Number of arcs

Visual Basic for Applications (VBA) Syntax

See [DisplayData::GetArcCount](#).

Example

[Get Weld Bead Symbol Data \(VBA\)](#)
[Get Weld Bead Symbol Data \(VB.NET\)](#)
[Get Weld Bead Symbol Data \(C#\)](#)

See Also

[IDisplayData Interface](#)
[IDisplayData Members](#)
[IDisplayData::GetArcAtIndex2 Method](#)
[IDisplayData::!GetArcAtIndex2 Method](#)

SOLIDWORKS API Help

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GetArrowHeadAtIndex Method (IDisplayData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IDisplayData Interface](#) : GetArrowHeadAtIndex Method (IDisplayData)

Obsolete. Superseded by [IDisplayData::GetArrowheadAtIndex2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function GetArrowHeadAtIndex(  
    ByVal Index As System.Integer  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IDisplayData  
Dim Index As System.Integer  
Dim value As System.Object  
  
value = instance.GetArrowHeadAtIndex(Index)
```

C#

```
System.Object GetArrowHeadAtIndex(  
    System.int Index  
)
```

C++/CLI

```
System.Object^ GetArrowHeadAtIndex(  
& System.int Index  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Index
Index of the desired arrow head where the index begins at zero

Return Value

Array of doubles (see **Remarks**)

 Visual Basic for Applications (VBA) Syntax

See [DisplayData::GetArrowHeadAtIndex](#).

 Remarks

The return value is the following array of doubles:
[*arrowHeadPt*[3], *arrowHeadDir*[3], *arrowHeadWidth*, *arrowHeadHeight*, *arrowHeadStyle*]
where:

<i>arrowHeadPt</i> [3]	XYZ arrow head tip location
<i>arrowHeadDir</i> [3]	XYZ arrow head direction
<i>arrowHeadWidth</i>	Arrow head width measured along the arrow head direction
<i>arrowHeadHeight</i>	Arrow head height
<i>arrowHeadStyle</i>	Arrow head style as defined in swArrowStyle_e

 See Also

[IDisplayData Interface](#)
[IDisplayData Members](#)
[IDisplayData::GetArrowHeadAtIndex Method](#)
[IDisplayData::GetArrowHeadCount Method](#)

SOLIDWORKS API Help [Send comments on this topic.](#)

GetArrowHeadAtIndex2 Method (IDisplayData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IDisplayData Interface](#) : GetArrowHeadAtIndex2 Method (IDisplayData)

Gets information on the specified arrow head.

 .NET Syntax

Visual Basic (Declaration)

```
Function GetArrowHeadAtIndex2( _  
    ByVal Index As System.Integer _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IDisplayData
Dim Index As System.Integer
Dim value As System.Object

value = instance.GetArrowHeadAtIndex2(Index)
```

C#

```
System.object GetArrowHeadAtIndex2(
    System.int Index
)
```

C++/CLI

```
System.Object^ GetArrowHeadAtIndex2(
    & System.int Index
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Index
Index of the desired arrow head where the index begins at zero

Return Value

Array of doubles (see **Remarks**)

 Visual Basic for Applications (VBA) Syntax

See [DisplayData::GetArrowHeadAtIndex2](#).

 Remarks

The return value is the following array of doubles:
[*arrowHeadPt[3]*, *arrowHeadDir[3]*, *arrowHeadWidth*, *arrowHeadHeight*, *arrowHeadStyle*, *arrowHeadNormal[3]*]
where:

Array member...	Contains...
<i>arrowHeadPt[3]</i>	XYZ coordinates of arrow head tip location
<i>arrowHeadDir[3]</i>	XYZ coordinates of arrow head direction
<i>arrowHeadWidth</i>	Arrow head width measured along the arrow head direction
<i>arrowHeadHeight</i>	Arrow head height
<i>arrowHeadStyle</i>	Arrow head style as defined in swArrowStyle_e
<i>arrowHeadNormal[3]</i>	XYZ coordinates of normal to the plane of the arrow head

 See Also

[IDisplayData Interface](#)
[IDisplayData Members](#)
[IDisplayData::GetArrowHeadCount Method \(\)](#)

 Availability

SOLIDWORKS 2015 FCS, Revision Number 23.0

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.NET Syntax

Visual Basic (Declaration)

```
Function GetArrowHeadCount() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IDisplayData  
Dim value As System.Integer  
  
value = instance.GetArrowHeadCount()
```

C#

```
System.int GetArrowHeadCount()
```

C++/CLI

```
System.int GetArrowHeadCount();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Number of arrow heads

Visual Basic for Applications (VBA) Syntax

See [DisplayData::GetArrowHeadCount](#).

See Also

[IDisplayData Interface](#)
[IDisplayData Members](#)
[IDisplayData::GetArrowHeadAtIndex Method](#)
[IDisplayData::IGetArrowHeadAtIndex Method](#)

SOLIDWORKS API Help

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GetEllipseAtIndex Method (IDisplayData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IDisplayData Interface](#) : GetEllipseAtIndex Method (IDisplayData)

Obsolete. Superseded by [IDisplayData::GetEllipseAtIndex2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function GetEllipseAtIndex( _  
    ByVal Index As System.Integer _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IDisplayData  
Dim Index As System.Integer  
Dim value As System.Object  
  
value = instance.GetEllipseAtIndex(Index)
```

C#

```
System.object GetEllipseAtIndex(  
    System.int Index  
)
```

C++/CLI

```
System.Object^ GetEllipseAtIndex(  
& System.int Index  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Index

Visual Basic for Applications (VBA) Syntax

See [DisplayData::GetEllipseAtIndex](#).

See Also

[IDisplayData Interface](#)
[IDisplayData Members](#)

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GetEllipseAtIndex2 Method (IDisplayData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IDisplayData Interface](#) : GetEllipseAtIndex2 Method (IDisplayData)

Gets information for the specified ellipse.

.NET Syntax

Visual Basic (Declaration)

```
Function GetEllipseAtIndex2( _  
    ByVal Index As System.Integer _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IDisplayData  
Dim Index As System.Integer  
Dim value As System.Object
```

```
value = instance.GetEllipseAtIndex2(Index)
```

C#

```
System.object GetEllipseAtIndex2(  
    System.int Index  
)
```

C++/CLI

```
System.Object^ GetEllipseAtIndex2(  
& System.int Index  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Index

Index of the desired ellipse where the index begins at zero

Return Value

Array of doubles (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [DisplayData::GetEllipseAtIndex2](#).

Remarks

The return value is the following array of doubles:

[*color*, *lineType*, *Unused*, *Unused*, *StartPt*[3], *EndPt*[3], *CenterPt*[3], *MajorPt*[3], *MinorPt*[3],]

where:

<i>color</i>	COLORREF returned as an integer. Return value could be 0 or -1 for default color.
<i>LineType</i>	Line type as defined in swLineTypes_e .
<i>StartPt</i> [3]	Array of 3 doubles (X,Y,Z) describing the ellipse start point.
<i>EndPt</i> [3]	Array of 3 doubles (X,Y,Z) describing the ellipse end point. If the ellipse is closed, then this will be the same point as the StartPt.
<i>CenterPt</i> [3]	Array of 3 doubles (X,Y,Z) describing the ellipse center point.
<i>MajorPt</i> [3]	Array of 3 doubles (X,Y,Z) describing a point on the ellipse and the major axis.
<i>MinorPt</i> [3]	Array of 3 doubles (X,Y,Z) describing a point on the ellipse and the minor axis.

See Also

[IDisplayData Interface](#)
[IDisplayData Members](#)
[IDisplayData::IGetEllipseAtIndex2 Method](#)
[IDisplayData::GetEllipseCount Method](#)

SOLIDWORKS API Help

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GetEllipseCount Method (IDisplayData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IDisplayData Interface](#) : GetEllipseCount Method (IDisplayData)

Gets the number of line ellipses.

.NET Syntax

Visual Basic (Declaration)

```
Function GetEllipseCount() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IDisplayData  
Dim value As System.Integer  
  
value = instance.GetEllipseCount()
```

C#

```
System.int GetEllipseCount()
```

C++/CLI

```
System.int GetEllipseCount();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Number of ellipses

Visual Basic for Applications (VBA) Syntax

See [DisplayData::GetEllipseCount](#).

See Also

[IDisplayData Interface](#)
[IDisplayData Members](#)
[IDisplayData::GetEllipseAtIndex2 Method](#)
[IDisplayData::IGetEllipseAtIndex2 Method](#)

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GetLineAtIndex Method (IDisplayData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IDisplayData Interface](#) : GetLineAtIndex Method (IDisplayData)

Obsolete. Superseded by [IDisplayData::GetLineAtIndex2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function GetLineAtIndex(  
    ByVal Index As System.Integer _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IDisplayData  
Dim Index As System.Integer  
Dim value As System.Object  
  
value = instance.GetLineAtIndex(Index)
```

C#

```
System.Object GetLineAtIndex(  
    System.int Index  
)
```

C++/CLI

```
System.Object^ GetLineAtIndex(  
& System.int Index  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Index

Visual Basic for Applications (VBA) Syntax

See [DisplayData::GetLineAtIndex](#).

See Also

[IDisplayData Interface](#)
[IDisplayData Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetLineAtIndex2 Method (IDisplayData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IDisplayData Interface](#) : GetLineAtIndex2 Method (IDisplayData)

Obsolete. Superseded by [DisplayData::GetLineAtIndex3](#).

.NET Syntax

Visual Basic (Declaration)

```
Function GetLineAtIndex2( _  
    ByVal Index As System.Integer _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IDisplayData  
Dim Index As System.Integer  
Dim value As System.Object  
  
value = instance.GetLineAtIndex2(Index)
```

C#

```
System.object GetLineAtIndex2(  
    System.int Index  
)
```

C++/CLI

```
System.Object^ GetLineAtIndex2(  
& System.int Index  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Index
Index of the desired line where the index begins at zero

Return Value

Array of doubles (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [DisplayData::GetLineAtIndex2](#).

Remarks

The return value is the following array of doubles:

[*color*, *lineType*, *Unused*, *Unused*, *startPt*[3], *endPt*[3]]

where:

<i>color</i>	COLORREF returned as an integer; return value can be 0 or -1 for default color
<i>lineType</i>	Line type as defined in swLineTypes_e
<i>startPt</i> [3]	x, y, z line start point
<i>endPt</i> [3]	x, y, z line end point

See Also

[IDisplayData Interface](#)
[IDisplayData Members](#)
[IDisplayData::IGetLineAtIndex2 Method](#)
[IDisplayData::GetLineCount Method](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

GetLineAtIndex3 Method (IDisplayData)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IDisplayData Interface](#) : GetLineAtIndex3 Method (IDisplayData)

Gets information for the specified line.

Visual Basic Syntax

Visual Basic (Declaration)

```
Function GetLineAtIndex3( _  
    ByVal Index As System.Integer _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IDisplayData  
Dim Index As System.Integer  
Dim value As System.Object  
  
value = instance.GetLineAtIndex3(Index)
```

C#

```
System.object GetLineAtIndex3(  
    System.int Index  
)
```

C++/CLI

```
System.Object^ GetLineAtIndex3(  
& System.int Index  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Index

Index of the desired line where the index begins at 0

Return Value

Array of doubles (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [DisplayData::GetLineAtIndex3](#).

Example

[Get Centerline Annotation Information \(VBA\)](#)
[Get Weld Bead Symbol Data \(VBA\)](#)
[Get Weld Bead Symbol Data \(VB.NET\)](#)
[Get Weld Bead Symbol Data \(C#\)](#)

Remarks

The return value is the following array of doubles:

[*color*, *lineType*, *lineStyle*, *lineWeight*, *startPt[3]*, *endPt[3]*]

where:

<i>color</i>	COLORREF returned as an integer; return value can be 0 or -1 for default color
<i>lineType</i>	Line type as defined in swLineTypes_e
<i>lineStyle</i>	Line style as defined in swLineStyles_e
<i>lineWeight</i>	Line thickness as defined in swLineWeights_e
<i>startPt[3]</i>	x, y, z line start point
<i>endPt[3]</i>	x, y, z line end point

See Also

[IDisplayData Interface](#)
[IDisplayData Members](#)
[IDisplayData::IGetLineAtIndex3 Method](#)
[IDisplayData::GetLineCount Method](#)

Availability

SOLIDWORKS SP3, Revision Number 16.3

SOLIDWORKS API Help

[Send comments](#) on this topic.

GetLineCount Method (IDisplayData)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IDisplayData Interface](#) : GetLineCount Method (IDisplayData)

Gets the number of lines.

.NET Syntax

Visual Basic (Declaration)

```
Function GetLineCount() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IDisplayData  
Dim value As System.Integer  
  
value = instance.GetLineCount()
```

C#

```
System.int GetLineCount()
```

C++/CLI

```
System.int GetLineCount();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Number of lines

Visual Basic for Applications (VBA) Syntax

See [DisplayData::GetLineCount](#).

Example

[Get the Number of Lines in Flat-pattern Drawing View's Boundary-box Sketch \(C#\)](#)
[Get the Number of Lines in Flat-pattern Drawing View's Boundary-box Sketch \(VB.NET\)](#)
[Get the Number of Lines in Flat-pattern Drawing View's Boundary-box Sketch \(VBA\)](#)
[Get Weld Bead Symbol Data \(VBA\)](#)
[Get Weld Bead Symbol Data \(VB.NET\)](#)
[Get Weld Bead Symbol Data \(C#\)](#)

See Also

[IDisplayData Interface](#)
[IDisplayData Members](#)
[IDisplayData::GetLineAtIndex2 Method](#)
[IDisplayData::IGetLineAtIndex2 Method](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

GetParabolaAtIndex Method (IDisplayData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IDisplayData Interface](#) : GetParabolaAtIndex Method (IDisplayData)

Gets information for the specified parabola.

 .NET Syntax

Visual Basic (Declaration)

```
Function GetParabolaAtIndex( _  
    ByVal Index As System.Integer _  
) As System.Object
```

Visual Basic (Usage)

```
Dim instance As IDisplayData  
Dim Index As System.Integer  
Dim value As System.Object  
  
value = instance.GetParabolaAtIndex(Index)
```

C#

```
System.object GetParabolaAtIndex(  
    System.int Index  
)
```

C++/CLI

```
System.Object^ GetParabolaAtIndex(  
& System.int Index  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Index
Index of parabola

Return Value

Array (see **Remarks**)

 Visual Basic for Applications (VBA) Syntax

See [DisplayData::GetParabolaAtIndex](#).

 Remarks

The return values are in an array of 16 doubles:
[*Color*, *LineType*, *LineStyleIndex*, *LineWeight*, *StartPt*[3], *EndPt*[3], *FocusPt*[3], *ApexPt*[3]]
where:

<i>Color</i>	COLORREF returned as an integer; return value can be 0 or -1 for the default color
<i>LineType</i>	Line type as defined in swLineTypes_e
<i>LineStyleIndex</i>	Index location of this line style inside SOLIDWORKS Line Style Manager
<i>LineWeight</i>	Integer value defining the line weight
<i>StartPt</i> [3]	Array of 3 doubles (X,Y,Z) describing the parabola start point
<i>EndPt</i> [3]	Array of 3 doubles (X,Y,Z) describing the parabola end point
<i>FocusPt</i> [3]	Array of 3 doubles (X,Y,Z) describing the parabola focus point
<i>ApexPt</i> [3]	Array of 3 doubles (X,Y,Z) describing the parabola apex point

This set of data repeats for each parabola in the view. The size of the array is (NumParabolas * 16). Use [IDisplayData::GetParabolaCount](#) to determine the number of parabolas.

See Also

[IDisplayData Interface](#)
[IDisplayData Members](#)
[IDisplayData::IGetParabolaAtIndex Method](#)
[IDisplayData::GetParabolaCount Method](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetParabolaCount Method (IDisplayData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IDisplayData Interface](#) : GetParabolaCount Method (IDisplayData)

Gets the number of parabolas.

.NET Syntax

Visual Basic (Declaration)

```
Function GetParabolaCount() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IDisplayData  
Dim value As System.Integer
```

```
value = instance.GetParabolaCount()
```

C#

```
System.int GetParabolaCount()
```

C++/CLI

```
System.int GetParabolaCount();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Number of parabolas

Visual Basic for Applications (VBA) Syntax

See [DisplayData::GetParabolaCount](#).

See Also

[IDisplayData Interface](#)
[IDisplayData Members](#)
[IDisplayData::GetParabolaAtIndex Method](#)
[IDisplayData::IGetParabolaAtIndex Method](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetPointAtIndex Method (IDisplayData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IDisplayData Interface](#) : GetPointAtIndex Method (IDisplayData)

Gets information about the specified point.

.NET Syntax

Visual Basic (Declaration)

```
Function GetPointAtIndex(  
    ByVal Index As System.Integer  
) As System.Object
```

Visual Basic (Usage)

```
Dim instance As IDisplayData
Dim Index As System.Integer
Dim value As System.Object

value = instance.GetPointAtIndex(Index)
```

C#

```
System.Object GetPointAtIndex(
    System.int Index
)
```

C++/CLI

```
System.Object^ GetPointAtIndex(
    & System.int Index
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Index
Index of the point

Return Value

Array (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [DisplayData::GetPointAtIndex](#).

Remarks

The return values are in an array of 7 doubles:

[*Color*, *LineType*, *LineStyleIndex*, *PointStyle*, *CircleFilled*, *X*, *Y*, *Z*]

where:

<i>Color</i>	COLORREF returned as an integer; return value can be 0 or -1 for the default color
<i>LineType</i>	Line type as defined in swLineTypes_e
<i>PointStyle</i>	Point style as defined in swPointStyle_e
<i>CircleFilled</i>	0 if the circle is filled, 1 if not
<i>X</i>	X coordinate of the specified point
<i>Y</i>	Y coordinate of the specified point
<i>Z</i>	Z coordiante of the specified point

Use [IDisplayData::GetPointCount](#) to determine the number of points.

See Also

[IDisplayData Interface](#)
[IDisplayData Members](#)
[IDisplayData::IGetPointAtIndex Method](#)

Availability

SOLIDWORKS 2009 FCS, Revision Number 17.0

Gets the number of points.

.NET Syntax

Visual Basic (Declaration)

```
Function GetPointCount() As System.Integer
```

Visual Basic (Usage)

 Copy Code

```
Dim instance As IDisplayData  
Dim value As System.Integer  
  
value = instance.GetPointCount()
```

C#

```
System.int GetPointCount()
```

C++/CLI

```
System.int GetPointCount();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Number of points

Visual Basic for Applications (VBA) Syntax

See [DisplayData::GetPointCount](#).

See Also

[IDisplayData Interface](#)
[IDisplayData Members](#)
[IDisplayData::GetPointAtIndex Method](#)
[IDisplayData::IGetPointAtIndex Method](#)

Availability

SOLIDWORKS 2009 FCS, Revision Number 17.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

GetPolygonAtIndex Method (IDisplayData)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IDisplayData Interface](#) : GetPolygonAtIndex Method (IDisplayData)

Gets information for the specified polygon.

.NET Syntax

Visual Basic (Declaration)

```
Function GetPolygonAtIndex( _  
    ByVal Index As System.Integer _  
) As System.Object
```

Visual Basic (Usage)

 Copy Code

```
Dim instance As IDisplayData  
Dim Index As System.Integer  
Dim value As System.Object  
  
value = instance.GetPolygonAtIndex(Index)
```

C#

```
System.object GetPolygonAtIndex(  
    System.int Index  
)
```

C++/CLI

```
System.Object^ GetPolygonAtIndex(  
& System.int Index  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Index

Index of the desired polygon where the index begins at zero

Return Value

Array of doubles (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [DisplayData::GetPolygonAtIndex](#).

Example

[Get Weld Bead Symbol Data \(VBA\)](#)
[Get Weld Bead Symbol Data \(VB.NET\)](#)
[Get Weld Bead Symbol Data \(C#\)](#)

Remarks

Format of return values is an array of doubles with the format:

[*Color*, *LineType*, *Unused*, *Unused*, *NumPolyPoints*, [x,y,z]]

where:

<i>Color</i>	Polygon color
<i>LineType</i>	Line type as defined in swLineTypes e. You can use this index value with IDrawingDoc::GetLineFontInfo2 and IDrawingDoc::GetLineFontName2
<i>NumPolyPoints</i>	Number of XYZ points found in the [x,y,z] return value
[x,y,z]	Array of NumPolyPoints points used to describe the polygon

Use [IDisplayData::GetPolygonSizeAtIndex](#) to determine the number of elements returned in this array.

See Also

[IDisplayData Interface](#)
[IDisplayData Members](#)
[IDisplayData::GetPolygonCount Method](#)
[IDisplayData::!GetPolygonAtIndex Method](#)
[IDisplayData::GetPolygonSizeAtIndex Method](#)

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GetPolygonCount Method (IDisplayData)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IDisplayData Interface](#) : GetPolygonCount Method (IDisplayData)

Gets the number of polygons.

.NET Syntax

Visual Basic (Declaration)

```
Function GetPolygonCount() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IDisplayData  
Dim value As System.Integer  
  
value = instance.GetPolygonCount()
```

C#

```
System.int GetPolygonCount()
```

C++/CLI

```
System.int GetPolygonCount();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Number of polygons

Visual Basic for Applications (VBA) Syntax

See [DisplayData::GetPolygonCount](#).

Example

[Get Weld Bead Symbol Data \(VBA\)](#)
[Get Weld Bead Symbol Data \(VB.NET\)](#)
[Get Weld Bead Symbol Data \(C#\)](#)

See Also

[IDisplayData Interface](#)
[IDisplayData Members](#)
[IDisplayData::GetPolygonAtIndex Method](#)
[IDisplayData::GetPolygonSizeAtIndex Method](#)
[IDisplayData::IGetPolygonAtIndex Method](#)

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GetPolygonSizeAtIndex Method (IDisplayData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IDisplayData Interface](#) : GetPolygonSizeAtIndex Method (IDisplayData)

Gets the number of array elements returned by [IDisplayData::GetPolygonAtIndex](#) and [IDisplayData::IGetPolygonAtIndex](#).

.NET Syntax

Visual Basic (Declaration)

```
Function GetPolygonSizeAtIndex( _  
    ByVal Index As System.Integer _  
) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IDisplayData  
Dim Index As System.Integer  
Dim value As System.Integer  
  
value = instance.GetPolygonSizeAtIndex(Index)
```

C#

```
System.int GetPolygonSizeAtIndex(  
    System.int Index  
)
```

C++/CLI

```
System.int GetPolygonSizeAtIndex(  
& System.int Index  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Index
Index for the polygon

Return Value

Number of array elements returned by [IDisplayData::GetPolygonAtIndex](#) and [IDisplayData::IGetPolygonAtIndex](#)

Visual Basic for Applications (VBA) Syntax

See [DisplayData::GetPolygonSizeAtIndex](#).

See Also

[IDisplayData Interface](#)
[IDisplayData Members](#)
[IDisplayData::GetPolygonCount Method](#)

Availability

SOLIDWORKS 2000 FCS, Revision Number 8.0

SOLIDWORKS API Help

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GetPolylineAtIndex Method (IDisplayData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IDisplayData Interface](#) : GetPolylineAtIndex Method (IDisplayData)

Obsolete. Superseded by [IDisplayData::GetPolylineAtIndex2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function GetPolylineAtIndex( _  
    ByVal Index As System.Integer _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IDisplayData  
Dim Index As System.Integer  
Dim value As System.Object  
  
value = instance.GetPolylineAtIndex(Index)
```

C#

```
System.object GetPolylineAtIndex(  
    System.int Index  
)
```

C++/CLI

```
System.Object^ GetPolylineAtIndex(  
& System.int Index  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Index

Visual Basic for Applications (VBA) Syntax

See [DisplayData::GetPolylineAtIndex](#).

See Also

[IDisplayData Interface](#)
[IDisplayData Members](#)

SOLIDWORKS API Help

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GetPolylineAtIndex2 Method (IDisplayData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IDisplayData Interface](#) : GetPolylineAtIndex2 Method (IDisplayData)

Gets information for the specified polyline.

.NET Syntax

Visual Basic (Declaration)

```
Function GetPolylineAtIndex2( _  
    ByVal Index As System.Integer _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IDisplayData  
Dim Index As System.Integer  
Dim value As System.Object
```

```
value = instance.GetPolylineAtIndex2(Index)
```

C#

```
System.object GetPolylineAtIndex2(  
    System.int Index  
)
```

C++/CLI

```
System.Object^ GetPolylineAtIndex2(  
& System.int Index  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Index

Index of the desired polyline where the index begins at zero

Return Value

Array of doubles (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [DisplayData::GetPolylineAtIndex2](#).

Remarks

Format of return values is an array of doubles with the format:

[*polyLineType*, *DataSize*, *Color*, *LineFont*, *Unused*, *Unused*, *NumPolyPoints*, [x,y,z]]

where:

<i>polyLineType</i>	Underlying geometry type: 0 - Polyline type 1 - Arc or Circle type
<i>DataSize</i>	Number of elements in the <i>GeomData</i> array. For Type = 0, this argument is 0.
<i>Color</i>	Polyline color.
<i>LineFont</i>	Polyline font information. You can use his value with IDrawingDoc::GetLineFontInfo2 and IDrawingDoc::GetLineFontName2 . This value is valid only if <i>LineStyle</i> is -1.
<i>NumPolyPoints</i>	Number of XYZ points found in the [x,y,z] return value.
[x,y,z]	Array of points used to describe the polyline. This array has <i>NumPolyPoints</i> points. This method returns an array for every polyline regardless of its type.

Use [IDisplayData::GetPolyLineSizeAtIndex2](#) to determine the number of elements returned in this array.

See Also

[IDisplayData Interface](#)
[IDisplayData Members](#)
[IDisplayData::GetPolyLineCount Method](#)
[IDisplayData::GetPolylineAtIndex2 Method](#)
[IDisplayData::GetPolylineSizeAtIndex2 Method](#)

SOLIDWORKS API Help

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GetPolyLineCount Method (IDisplayData)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IDisplayData Interface](#) : GetPolyLineCount Method (IDisplayData)

Gets the number of polylines.

.NET Syntax

Visual Basic (Declaration)

```
Function GetPolyLineCount() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IDisplayData  
Dim value As System.Integer
```

```
value = instance.GetPolyLineCount()
```

C#

```
System.int GetPolyLineCount()
```

C++/CLI

```
System.int GetPolyLineCount();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Number of polylines

Visual Basic for Applications (VBA) Syntax

See [DisplayData::GetPolyLineCount](#).

Example

[Get Weld Bead Symbol Data \(VBA\)](#)
[Get Weld Bead Symbol Data \(VB.NET\)](#)
[Get Weld Bead Symbol Data \(C#\)](#)

See Also

[IDisplayData Interface](#)
[IDisplayData Members](#)
[IDisplayData::GetPolylineAtIndex2 Method](#)
[IDisplayData::GetPolylineSizeAtIndex2 Method](#)
[IDisplayData::IGetPolylineAtIndex2 Method](#)

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GetPolylineSizeAtIndex Method (IDisplayData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IDisplayData Interface](#) : GetPolylineSizeAtIndex Method (IDisplayData)

Obsolete. Superseded by [IDisplayData::GetPolylineSizeAtIndex2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function GetPolylineSizeAtIndex( _  
    ByVal Index As System.Integer _  
) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IDisplayData  
Dim Index As System.Integer  
Dim value As System.Integer
```

```
value = instance.GetPolylineSizeAtIndex(Index)
```

C#

```
System.int GetPolylineSizeAtIndex(  
    System.int Index  
)
```

C++/CLI

```
System.int GetPolylineSizeAtIndex(  
    & System.int Index  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Index

Visual Basic for Applications (VBA) Syntax

See [DisplayData::GetPolylineSizeAtIndex](#).

See Also

[IDisplayData Interface](#)
[IDisplayData Members](#)

GetPolylineSizeAtIndex2 Method (IDisplayData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IDisplayData Interface](#) : GetPolylineSizeAtIndex2 Method (IDisplayData)

Gets the number of array elements returned by [IDisplayData::GetPolylineAtIndex2](#) and [IDisplayData::IGetPolylineAtIndex2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function GetPolylineSizeAtIndex2( _  
    ByVal Index As System.Integer _  
) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IDisplayData  
Dim Index As System.Integer  
Dim value As System.Integer
```

```
value = instance.GetPolylineSizeAtIndex2(Index)
```

C#

```
System.int GetPolylineSizeAtIndex2(  
    System.int Index  
)
```

C++/CLI

```
System.int GetPolylineSizeAtIndex2(  
&    System.int Index  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Index

Index of the desired polyline where the index begins at zero

Return Value

Number array elements returned by [IDisplayData::GetPolylineAtIndex2](#) and [IDisplayData::IGetPolylineAtIndex2](#)

Visual Basic for Applications (VBA) Syntax

See [DisplayData::GetPolylineSizeAtIndex2](#).

See Also

[IDisplayData Interface](#)

[IDisplayData Members](#)

[IDisplayData::GetPolyLineCount Method](#)

GetTextAngleAtIndex Method (IDisplayData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IDisplayData Interface](#) : GetTextAngleAtIndex Method (IDisplayData)

Gets the text angle for the specified piece of text.

.NET Syntax

Visual Basic (Declaration)

```
Function GetTextAngleAtIndex( _  
    ByVal Index As System.Integer _
```

```
) As System.Double
```

Visual Basic (Usage)



```
Dim instance As IDisplayData  
Dim Index As System.Integer  
Dim value As System.Double  
  
value = instance.GetTextAngleAtIndex(Index)
```

C#

```
System.double GetTextAngleAtIndex(  
    System.int Index  
)
```

C++/CLI

```
System.double GetTextAngleAtIndex(  
&    System.int Index  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Index

Zero-based index of the desired piece of text

Return Value

Text angle for the specified text in radians measured CCW from the X axis

Visual Basic for Applications (VBA) Syntax

See [DisplayData::GetTextAngleAtIndex](#).

See Also

[IDisplayData Interface](#)
[IDisplayData Members](#)
[IDisplayData::IDisplayData::GetTextAtIndex Method](#)
[IDisplayData::IDisplayData::GetTextCount Method](#)
[IDisplayData::GetTextFontAtIndex Method](#)
[IDisplayData::GetTextHeightAtIndex Method](#)
[IDisplayData::GetTextInBoxHeightAtIndex Method](#)
[IDisplayData::GetTextInBoxStyleAtIndex Method](#)
[IDisplayData::GetTextInBoxWidthAtIndex Method](#)
[IDisplayData::GetTextInvertAtIndex Method](#)
[IDisplayData::GetTextLineSpacingAtIndex Method](#)
[IDisplayData::GetTextPositionAtIndex Method](#)
[IDisplayData::GetTextRefPositionAtIndex Method](#)
[IDisplayData::IGetTextPositionAtIndex Method](#)
[IDisplayData::GetTextPlaneAtIndex Method \(\)](#)

SOLIDWORKS API Help

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GetTextAtIndex Method (IDisplayData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IDisplayData Interface](#) : GetTextAtIndex Method (IDisplayData)

Gets the specified text string.

.NET Syntax

Visual Basic (Declaration)

```
Function GetTextAtIndex( _  
    ByVal Index As System.Integer _  
) As System.String
```

Visual Basic (Usage)



```
Dim instance As IDisplayData  
Dim Index As System.Integer
```

```
Dim value As System.String

value = instance.GetTextAtIndex(Index)
```

C#

```
System.string GetTextAtIndex(
    System.int Index
)
```

C++/CLI

```
System.String^ GetTextAtIndex(
    & System.int Index
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Index
Zero-based index of the desired piece of text

Return Value

Text string for the specified piece of text

Visual Basic for Applications (VBA) Syntax

See [DisplayData::GetTextAtIndex](#).

See Also

[IDisplayData Interface](#)
[IDisplayData Members](#)
[IDisplayData::GetTextAngleAtIndex Method](#)
[IDisplayData::GetTextCount Method](#)
[IDisplayData::GetTextFontAtIndex Method](#)
[IDisplayData::GetTextHeightAtIndex Method](#)
[IDisplayData::GetTextInBoxHeightAtIndex Method](#)
[IDisplayData::GetTextInBoxStyleAtIndex Method](#)
[IDisplayData::GetTextInBoxWidthAtIndex Method](#)
[IDisplayData::GetTextInvertAtIndex Method](#)
[IDisplayData::GetTextLineSpacingAtIndex Method](#)
[IDisplayData::GetTextPositionAtIndex Method](#)
[IDisplayData::GetTextRefPositionAtIndex Method](#)
[IDisplayData::IGetTextPositionAtIndex Method](#)
[IDisplayData::GetTextPlaneAtIndex Method \(\)](#)

SOLIDWORKS API Help

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GetTextCount Method (IDisplayData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IDisplayData Interface](#) : GetTextCount Method (IDisplayData)

Gets the number of text items.

.NET Syntax

Visual Basic (Declaration)

```
Function GetTextCount() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IDisplayData
Dim value As System.Integer

value = instance.GetTextCount()
```

C#

```
System.int GetTextCount()
```

C++/CLI

```
System.int GetTextCount();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Number of text items

Visual Basic for Applications (VBA) Syntax

See [DisplayData::GetTextCount](#).

See Also

[IDisplayData Interface](#)
[IDisplayData Members](#)
[IDisplayData::GetTextAngleAtIndex Method](#)
[IDisplayData::GetTextAtIndex Method](#)
[IDisplayData::GetTextFontAtIndex Method](#)
[IDisplayData::GetTextHeightAtIndex Method](#)
[IDisplayData::GetTextInBoxHeightAtIndex Method](#)
[IDisplayData::GetTextInBoxStyleAtIndex Method](#)
[IDisplayData::GetTextInBoxWidthAtIndex Method](#)
[IDisplayData::GetTextInvertAtIndex Method](#)
[IDisplayData::GetTextLineSpacingAtIndex Method](#)
[IDisplayData::GetTextPositionAtIndex Method](#)
[IDisplayData::GetTextRefPositionAtIndex Method](#)
[IDisplayData::IGetTextPositionAtIndex Method](#)
[IDisplayData::GetTextPlaneAtIndex Method \(\)](#)

SOLIDWORKS API Help

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GetTextFontAtIndex Method (IDisplayData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IDisplayData Interface](#) : GetTextFontAtIndex Method (IDisplayData)

Gets the specified text font.

.NET Syntax

Visual Basic (Declaration)

```
Function GetTextFontAtIndex( _  
    ByVal Index As System.Integer _  
) As System.String
```

Visual Basic (Usage)



```
Dim instance As IDisplayData  
Dim Index As System.Integer  
Dim value As System.String  
  
value = instance.GetTextFontAtIndex(Index)
```

C#

```
System.string GetTextFontAtIndex(  
    System.int Index  
)
```

C++/CLI

```
System.String^ GetTextFontAtIndex(  
    & System.int Index  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Index

Zero-based index of the desired piece of text

Return Value

Font for the specified piece of text

Visual Basic for Applications (VBA) Syntax

See [DisplayData::GetTextFontAtIndex](#).

See Also

[IDisplayData Interface](#)
[IDisplayData Members](#)
[IDisplayData::GetTextCount Method](#)
[IDisplayData::GetTextAngleAtIndex Method](#)
[IDisplayData::GetTextAtIndex Method](#)
[IDisplayData::GetTextHeightAtIndex Method](#)
[IDisplayData::GetTextInBoxHeightAtIndex Method](#)
[IDisplayData::GetTextInBoxStyleAtIndex Method](#)
[IDisplayData::GetTextInBoxWidthAtIndex Method](#)
[IDisplayData::GetTextInvertAtIndex Method](#)
[IDisplayData::GetTextLineSpacingAtIndex Method](#)
[IDisplayData::GetTextPositionAtIndex Method](#)
[IDisplayData::IGetTextPositionAtIndex Method](#)
[IDisplayData::GetTextRefPositionAtIndex Method](#)
[IDisplayData::GetTextPlaneAtIndex Method \(\)](#)

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GetTextHeightAtIndex Method (IDisplayData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IDisplayData Interface](#) : GetTextHeightAtIndex Method (IDisplayData)

Gets the text height for the specified piece of text .

.NET Syntax

Visual Basic (Declaration)

```
Function GetTextHeightAtIndex( _  
    ByVal Index As System.Integer _  
) As System.Double
```

Visual Basic (Usage)



```
Dim instance As IDisplayData  
Dim Index As System.Integer  
Dim value As System.Double  
  
value = instance.GetTextHeightAtIndex(Index)
```

C#

```
System.double GetTextHeightAtIndex(  
    System.int Index  
)
```

C++/CLI

```
System.double GetTextHeightAtIndex(  
&    System.int Index  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Index

Zero-based index of the desired piece of text

Return Value

Text height for the specified piece of text in meters

Visual Basic for Applications (VBA) Syntax

See [DisplayData::GetTextHeightAtIndex](#).

See Also

[IDisplayData Interface](#)
[IDisplayData Members](#)
[IDisplayData::GetTextAngleAtIndex Method](#)
[IDisplayData::GetTextAtIndex Method](#)
[IDisplayData::GetTextCount Method](#)
[IDisplayData::GetTextFontAtIndex Method](#)
[IDisplayData::GetTextInBoxHeightAtIndex Method](#)
[IDisplayData::GetTextInBoxStyleAtIndex Method](#)
[IDisplayData::GetTextInBoxWidthAtIndex Method](#)
[IDisplayData::GetTextInvertAtIndex Method](#)
[IDisplayData::GetTextLineSpacingAtIndex Method](#)
[IDisplayData::GetTextPositionAtIndex Method](#)
[IDisplayData::GetTextRefPositionAtIndex Method](#)
[IDisplayData::!GetTextPositionAtIndex Method](#)
[IDisplayData::GetTextPlaneAtIndex Method \(\)](#)

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GetTextInBoxHeightAtIndex Method (IDisplayData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IDisplayData Interface](#) : GetTextInBoxHeightAtIndex Method (IDisplayData)

Gets the text-in-box height of the specified text and table cell in a table annotation.

.NET Syntax

Visual Basic (Declaration)

```
Function GetTextInBoxHeightAtIndex( _  
    ByVal Index As System.Integer _  
) As System.Double
```

Visual Basic (Usage)



```
Dim instance As IDisplayData  
Dim Index As System.Integer  
Dim value As System.Double
```

```
value = instance.GetTextInBoxHeightAtIndex(Index)
```

C#

```
System.double GetTextInBoxHeightAtIndex(  
    System.int Index  
)
```

C++/CLI

```
System.double GetTextInBoxHeightAtIndex(  
&    System.int Index  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Index

Zero-based index of the text and table cell to examine

Return Value

Height of the text-in-box in system units

Visual Basic for Applications (VBA) Syntax

See [DisplayData::GetTextInBoxHeightAtIndex](#).

Remarks

Text-in-box refers to a fixed-size area slightly smaller than the table cell.

If [IDisplayData::GetTextInBoxStyleAtIndex](#) returns swTextInBoxStyleNone, then this method returns 0 as well.

See Also

[IDisplayData Interface](#)
[IDisplayData Members](#)
[IDisplayData::GetTextAngleAtIndex Method](#)
[IDisplayData::GetTextAtIndex Method](#)
[IDisplayData::GetTextCount Method](#)
[IDisplayData::GetTextFontAtIndex Method](#)
[IDisplayData::GetTextHeightAtIndex Method](#)
[IDisplayData::GetTextInBoxStyleAtIndex Method](#)
[IDisplayData::GetTextInBoxWidthAtIndex Method](#)
[IDisplayData::GetTextInvertAtIndex Method](#)
[IDisplayData::GetTextLineSpacingAtIndex Method](#)
[IDisplayData::GetTextPositionAtIndex Method](#)
[IDisplayData::GetTextRefPositionAtIndex Method](#)
[IDisplayData::!GetTextPositionAtIndex Method](#)
[IDisplayData::GetTextPlaneAtIndex Method \(\)](#)

Availability

SOLIDWORKS 2004 FCS, Revision Number 12.0

SOLIDWORKS API Help

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GetTextInBoxStyleAtIndex Method (IDisplayData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IDisplayData Interface](#) : GetTextInBoxStyleAtIndex Method (IDisplayData)

Gets the text-in-box style of the specified text and table cell in a table annotation.

.NET Syntax

Visual Basic (Declaration)

```
Function GetTextInBoxStyleAtIndex( _  
    ByVal Index As System.Integer _  
) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IDisplayData  
Dim Index As System.Integer  
Dim value As System.Integer
```

```
value = instance.GetTextInBoxStyleAtIndex(Index)
```

C#

```
System.int GetTextInBoxStyleAtIndex(  
    System.int Index  
)
```

C++/CLI

```
System.int GetTextInBoxStyleAtIndex(  
&    System.int Index  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Index

Zero-based index of the text and table cell to examine

Return Value

Text-in-box style as defined by [swTextInBoxStyle_e](#)

Visual Basic for Applications (VBA) Syntax

See [DisplayData::GetTextInBoxStyleAtIndex](#).

Remarks

Text-in-box refers to a fixed-size area slightly smaller than the table cell.

See Also

[IDisplayData Interface](#)
[IDisplayData Members](#)
[DisplayData::GetTextAngleAtIndex Method](#)
[DisplayData::GetTextAtIndex Method](#)
[DisplayData::GetTextCount Method](#)
[DisplayData::GetTextFontAtIndex Method](#)
[DisplayData::GetTextHeightAtIndex Method](#)
[DisplayData::GetTextInBoxHeightAtIndex Method](#)
[DisplayData::GetTextInBoxWidthAtIndex Method](#)
[DisplayData::GetTextInvertAtIndex Method](#)
[DisplayData::GetTextLineSpacingAtIndex Method](#)
[DisplayData::GetTextPositionAtIndex Method](#)
[DisplayData::GetTextRefPositionAtIndex Method](#)
[DisplayData::IGetTextPositionAtIndex Method](#)
[IDisplayData::GetTextPlaneAtIndex Method \(\)](#)

Availability

SOLIDWORKS 2004 FCS, Revision Numb4er 12.0

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GetTextInBoxWidthAtIndex Method (IDisplayData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IDisplayData Interface](#) : GetTextInBoxWidthAtIndex Method (IDisplayData)

Gets the text-in-box width of the specified text and table cell in the table annotation.

.NET Syntax

Visual Basic (Declaration)

```
Function GetTextInBoxWidthAtIndex( _  
    ByVal Index As System.Integer _  
) As System.Double
```

Visual Basic (Usage)



```
Dim instance As IDisplayData  
Dim Index As System.Integer  
Dim value As System.Double  
  
value = instance.GetTextInBoxWidthAtIndex(Index)
```

C#

```
System.double GetTextInBoxWidthAtIndex(  
    System.int Index  
)
```

C++/CLI

```
System.double GetTextInBoxWidthAtIndex(  
&    System.int Index  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Index
Zero-based index of the text and table cell to examine

Return Value

Width of the text-in-box in system units

Visual Basic for Applications (VBA) Syntax

See [DisplayData::GetTextInBoxWidthAtIndex](#).

Remarks

Text-in-box refers to a fixed-size area slightly smaller than the table cell.

If [IDisplayData::GetTextInBoxStyleAtIndex](#) returns swTextInBoxStyleNone, then this method returns 0 as well.

See Also

[IDisplayData Interface](#)
[IDisplayData Members](#)
[DisplayData::IGetTextAngleAtIndex Method](#)
[DisplayData::IGetTextAtIndex Method](#)
[DisplayData::IGetTextCount Method](#)
[DisplayData::IGetTextFontAtIndex Method](#)
[DisplayData::IGetTextHeightAtIndex Method](#)
[DisplayData::IGetTextInBoxHeightAtIndex Method](#)
[DisplayData::IGetTextInBoxStyleAtIndex Method](#)
[DisplayData::IGetTextInvertAtIndex Method](#)
[DisplayData::IGetTextLineSpacingAtIndex Method](#)
[DisplayData::IGetTextPositionAtIndex Method](#)
[DisplayData::IGetTextRefPositionAtIndex Method](#)
[DisplayData::IGetTextPositionAtIndex Method](#)
[IDisplayData::GetTextPlaneAtIndex Method \(\)](#)

Availability

SOLIDWORKS 2004 FCS, Revision Number 12.0

SOLIDWORKS API Help

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GetTextInvertAtIndex Method (IDisplayData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IDisplayData Interface](#) : GetTextInvertAtIndex Method (IDisplayData)

Gets the specified text's invert flag.

.NET Syntax

Visual Basic (Declaration)

```
Function GetTextInvertAtIndex( _  
    ByVal Index As System.Integer _  
) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IDisplayData  
Dim Index As System.Integer  
Dim value As System.Integer  
  
value = instance.GetTextInvertAtIndex(Index)
```

C#

```
System.int GetTextInvertAtIndex(  
    System.int Index  
)
```

C++/CLI

```
System.int GetTextInvertAtIndex(  
& System.int Index  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Index

Zero-based index of the desired piece of text

Return Value

Text item invert flag

Visual Basic for Applications (VBA) Syntax

See [DisplayData::GetTextInvertAtIndex](#).

Remarks

The invert flag specifies whether the text is mirrored (reflected) about the X axis. Any reflection is applied after text rotation.

See Also

[IDisplayData Interface](#)
[IDisplayData Members](#)
[IDisplayData::GetTextAngleAtIndex Method](#)
[IDisplayData::GetTextAtIndex Method](#)
[IDisplayData::GetTextCount Method](#)
[IDisplayData::GetTextFontAtIndex Method](#)
[IDisplayData::GetTextHeightAtIndex Method](#)
[IDisplayData::GetTextInBoxHeightAtIndex Method](#)
[IDisplayData::GetTextInBoxStyleAtIndex Method](#)
[IDisplayData::GetTextInBoxWidthAtIndex Method](#)
[IDisplayData::GetTextLineSpacingAtIndex Method](#)
[IDisplayData::GetTextPositionAtIndex Method](#)
[IDisplayData::GetTextRefPositionAtIndex Method](#)
[IDisplayData::GetTextPositionAtIndex Method](#)
[IDisplayData::GetTextPlaneAtIndex Method \(\)](#)

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GetTextLineSpacingAtIndex Method (IDisplayData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IDisplayData Interface](#) : [GetTextLineSpacingAtIndex Method \(IDisplayData\)](#)

Gets the line spacing for this text.

.NET Syntax

Visual Basic (Declaration)

```
Function GetTextLineSpacingAtIndex( _  
    ByVal Index As System.Integer _  
) As System.Double
```

Visual Basic (Usage)



```
Dim instance As IDisplayData  
Dim Index As System.Integer  
Dim value As System.Double  
  
value = instance.GetTextLineSpacingAtIndex(Index)
```

C#

```
System.double GetTextLineSpacingAtIndex(  
    System.int Index  
)
```

C++/CLI

```
System.double GetTextLineSpacingAtIndex(  
&    System.int Index  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Index

Zero-based index of the text

Return Value

Line spacing

Visual Basic for Applications (VBA) Syntax

See [DisplayData::GetTextLineSpacingAtIndex](#).

Remarks

Because display data can have multiple pieces of text, you must specify the index value of the desired text.

See Also

[IDisplayData Interface](#)
[IDisplayData Members](#)
[IDisplayData::GetTextAngleAtIndex Method](#)
[IDisplayData::GetTextAtIndex Method](#)
[IDisplayData::GetTextCount Method](#)
[IDisplayData::GetTextFontAtIndex Method](#)
[IDisplayData::GetTextHeightAtIndex Method](#)
[IDisplayData::GetTextInBoxHeightAtIndex Method](#)
[IDisplayData::GetTextInBoxStyleAtIndex Method](#)
[IDisplayData::GetTextInBoxWidthAtIndex Method](#)
[IDisplayData::GetTextInvertAtIndex Method](#)
[IDisplayData::GetTextPositionAtIndex Method](#)
[IDisplayData::GetTextRefPositionAtIndex Method](#)
[IDisplayData::!GetTextPositionAtIndex Method](#)
[IDisplayData::GetTextPlaneAtIndex Method \(\)](#)

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GetTextPlaneAtIndex Method (IDisplayData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IDisplayData Interface](#) : GetTextPlaneAtIndex Method (IDisplayData)

Gets the rotation matrix of the specified text relative to the X-Y plane of the model.

.NET Syntax

Visual Basic (Declaration)

```
Function GetTextPlaneAtIndex( _  
    ByVal Index As System.Integer _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IDisplayData  
Dim Index As System.Integer  
Dim value As System.Object  
  
value = instance.GetTextPlaneAtIndex(Index)
```

C#

```
System.object GetTextPlaneAtIndex(  
    System.int Index  
)
```

C++/CLI

```
System.Object^ GetTextPlaneAtIndex(  
&    System.int Index  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Index

Zero-based index of the text whose plane rotation matrix to retrieve (see **Remarks**)

Return Value

Array of nine doubles of the text plane rotation matrix (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [DisplayData::GetTextPlaneAtIndex](#).

Example

'VBA

'Preconditions:

- '1. Open a part with three datum annotations in different planes.
- '2. Open the Immediate window.

'Postconditions:

'For each datum annotation:

- ' 1. Prints the text plane rotation matrix.
- ' 2. Calculates the X-axis of the text plane rotated by the text angle in the XY plane.
- ' 3. Calculates the cross product of the X- and Z-axis to obtain the new Y-axis.
- ' 4. Prints the new text plane rotation matrix.

'=====

Option Explicit

```
Dim swApp As SldWorks.SldWorks
Dim aDoc As ModelDoc2
Dim aExt As ModelDocExtension
Dim boolstatus As Boolean
Dim annotVar As Variant
Dim i As Integer
Dim j As Integer
Dim k As Integer
```

Sub main()

```
Set swApp = Application.SldWorks
Set aDoc = swApp.ActiveDoc
Set aExt = aDoc.Extension
```

```
annotVar = aExt.GetAnnotations()
If Not IsEmpty(annotVar) Then
```

```
    For i = 0 To UBound(annotVar)
        Dim aAnnot As Annotation
        Set aAnnot = annotVar(i)
```

```
        Dim swDispData As SldWorks.DisplayData
        Set swDispData = aAnnot.GetDisplayData
```

```
        Debug.Print "Number of texts in annotation: " & swDispData.GetTextCount()
```

```
        For j = 0 To swDispData.GetTextCount()
```

```
            Dim vPlane As Variant
            vPlane = swDispData.GetTextPlaneAtIndex(j)
```

```
            If Not (IsEmpty(vPlane)) Then
                Debug.Print "Text : " & swDispData.GetTextAtIndex(j)
                ' Normalize very small positive/negative plane array values to 0
                For k = 0 To 8
                    If vPlane(k) < 0 Then
                        If (vPlane(k) > -0.000001) Then
                            vPlane(k) = 0
                        End If
                    ElseIf (vPlane(k) < 0.000001) Then
                        vPlane(k) = 0
                    End If
                Next k
```

```
                Debug.Print "Normalized text plane rotation matrix: "
                Debug.Print "" & vPlane(0) & " " & vPlane(1) & " " & vPlane(2)
                Debug.Print "" & vPlane(3) & " " & vPlane(4) & " " & vPlane(5)
                Debug.Print "" & vPlane(6) & " " & vPlane(7) & " " & vPlane(8)
```

```
                Dim xAxis As Variant
                Dim yAxis As Variant
                Dim zAxis As Variant
```

```
                ' The ordering of the array returned by IDisplayData::GetTextPlaneAtIndex differs from other APIs
                xAxis = Array(vPlane(0), vPlane(3), vPlane(6))
                yAxis = Array(vPlane(1), vPlane(4), vPlane(7))
```

```

zAxis = Array(vPlane(2), vPlane(5), vPlane(8))

Dim angle As Double
angle = swDispData.GetTextAngleAtIndex(j)

' Rotate xAxis by angle in X-Y plane
Dim xAxisNew As Variant
xAxisNew = Array(Cos(angle) * xAxis(0) + Sin(angle) * yAxis(0), Cos(angle) * xAxis(1) + Sin(angle) * yAxis(1), Cos(angle) * xAxis(2) + Sin(angle) * yAxis(2))

' Since rotation in the X-Y plane does not change the normal (Z plane),
' take the cross product of X- and Z-axis arrays to get the new Y-axis array (Y = Z CROSS X)
yAxis = Array(zAxis(1) * xAxisNew(2) - zAxis(2) * xAxisNew(1), -(zAxis(0) * xAxisNew(2) - zAxis(2) * xAxisNew(0)), zAxis(0) * xAxisNew(1) - zAxis(1) * xAxisNew(0))

' Normalize very small positive/negative axis coordinates to 0
For k = 0 To 2
    If xAxisNew(k) < 0 Then
        If (xAxisNew(k) > -0.000001) Then
            xAxisNew(k) = 0
        End If
    ElseIf (xAxisNew(k) < 0.000001) Then
        xAxisNew(k) = 0
    End If
Next k

For k = 0 To 2
    If yAxis(k) < 0 Then
        If (yAxis(k) > -0.000001) Then
            yAxis(k) = 0
        End If
    ElseIf (yAxis(k) < 0.000001) Then
        yAxis(k) = 0
    End If
Next k

For k = 0 To 2
    If zAxis(k) < 0 Then
        If (zAxis(k) > -0.000001) Then
            zAxis(k) = 0
        End If
    ElseIf (zAxis(k) < 0.000001) Then
        zAxis(k) = 0
    End If
Next k

Debug.Print "Normalized text plane matrix after rotation:"
Debug.Print "" & xAxisNew(0) & " " & xAxisNew(1) & " " & xAxisNew(2)
Debug.Print "" & yAxis(0) & " " & yAxis(1) & " " & yAxis(2)
Debug.Print "" & zAxis(0) & " " & zAxis(1) & " " & zAxis(2)

End If
Debug.Print " "
Next j
Next i
End If
End Sub

```

Remarks

Use [IDisplayData::GetTextCount](#) to specify Index.

The nine values of the returned rotation matrix (coord(*i*)) are ordered differently from other plane APIs. If you need to manipulate the text plane axes as is shown in the Example section, use the following assignments:

X-axis: coord(0), coord(3), coord(6)

Y-axis: coord(1), coord(4), coord(7)

Z-axis: coord(2), coord(5), coord(8)

See Also

[IDisplayData Interface](#)
[IDisplayData Members](#)
[IAnnotation::GetPlane Method \(\)](#)
[IDisplayData::GetTextAngleAtIndex Method \(\)](#)
[IDisplayData::GetTextAtIndex Method \(\)](#)
[IDisplayData::GetTextFontAtIndex Method \(\)](#)
[IDisplayData::GetTextHeightAtIndex Method \(\)](#)

[IDisplayData::GetTextInBoxHeightAtIndex Method \(\)](#)
[IDisplayData::GetTextInBoxStyleAtIndex Method \(\)](#)
[IDisplayData::GetTextInBoxWidthAtIndex Method \(\)](#)
[IDisplayData::GetTextInvertAtIndex Method \(\)](#)
[IDisplayData::GetTextLineSpacingAtIndex Method \(\)](#)
[IDisplayData::GetTextPositionAtIndex Method \(\)](#)
[IDisplayData::GetTextRefPositionAtIndex Method \(\)](#)

Availability

SOLIDWORKS 2019 SP01, Revision Number 27.1

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GetTextPositionAtIndex Method (IDisplayData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IDisplayData Interface](#) : GetTextPositionAtIndex Method (IDisplayData)

Gets the text item offset relative to the note text point.

.NET Syntax

Visual Basic (Declaration)

```
Function GetTextPositionAtIndex( _  
    ByVal Index As System.Integer _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IDisplayData  
Dim Index As System.Integer  
Dim value As System.Object
```

```
value = instance.GetTextPositionAtIndex(Index)
```

C#

```
System.object GetTextPositionAtIndex(  
    System.int Index  
)
```

C++/CLI

```
System.Object^ GetTextPositionAtIndex(  
& System.int Index  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Index

Zero-based index of the desired piece of text

Return Value

Array of doubles (**Remarks**)

Visual Basic for Applications (VBA) Syntax

See [DisplayData::GetTextPositionAtIndex](#).

Remarks

The return value is the following array of doubles:

[*textPtX*, *textPtY*, *textPtZ*]

where these text position values are actually offset values from the origin of this display data.

See Also

[IDisplayData Interface](#)
[IDisplayData Members](#)
[IDisplayData::GetTextAngleAtIndex Method](#)
[IDisplayData::GetTextAtIndex Method](#)
[IDisplayData::GetTextCount Method](#)
[IDisplayData::GetTextFontAtIndex Method](#)
[IDisplayData::GetTextHeightAtIndex Method](#)
[IDisplayData::GetTextInBoxHeightAtIndex Method](#)
[IDisplayData::GetTextInBoxStyleAtIndex Method](#)
[IDisplayData::GetTextInBoxWidthAtIndex Method](#)
[IDisplayData::GetTextInvertAtIndex Method](#)
[IDisplayData::GetTextLineSpacingAtIndex Method](#)
[IDisplayData::GetTextRefPositionAtIndex Method](#)
[IDisplayData::GetTextPositionAtIndex Method](#)
[IDisplayData::GetTextPlaneAtIndex Method \(\)](#)

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GetTextRefPositionAtIndex Method (IDisplayData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IDisplayData Interface](#) : GetTextRefPositionAtIndex Method (IDisplayData)

Gets the specified text item reference position.

.NET Syntax

Visual Basic (Declaration)

```
Function GetTextRefPositionAtIndex( _  
    ByVal Index As System.Integer _  
) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IDisplayData  
Dim Index As System.Integer  
Dim value As System.Integer
```

```
value = instance.GetTextRefPositionAtIndex(Index)
```

C#

```
System.int GetTextRefPositionAtIndex(  
    System.int Index  
)
```

C++/CLI

```
System.int GetTextRefPositionAtIndex(  
& System.int Index  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Index

Zero-based index of the desired piece of text

Return Value

Reference position of the specified text item as defined in [swTextPosition_e](#)

Visual Basic for Applications (VBA) Syntax

See [DisplayData::GetTextRefPositionAtIndex](#).

See Also

[IDisplayData Interface](#)
[IDisplayData Members](#)
[IDisplayData::GetTextAngleAtIndex Method](#)
[IDisplayData::GetTextAtIndex Method](#)
[IDisplayData::GetTextCount Method](#)
[IDisplayData::GetTextFontAtIndex Method](#)
[IDisplayData::GetTextHeightAtIndex Method](#)
[IDisplayData::GetTextInBoxHeightAtIndex Method](#)
[IDisplayData::GetTextInBoxStyleAtIndex Method](#)
[IDisplayData::GetTextInBoxWidthAtIndex Method](#)
[IDisplayData::GetTextInvertAtIndex Method](#)
[IDisplayData::GetTextLineSpacingAtIndex Method](#)
[IDisplayData::GetTextPositionAtIndex Method](#)
[IDisplayData::!GetTextPositionAtIndex Method](#)
[IDisplayData::GetTextPlaneAtIndex Method \(\)](#)

SOLIDWORKS API Help

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GetTriangleAtIndex Method (IDisplayData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IDisplayData Interface](#) : GetTriangleAtIndex Method (IDisplayData)

Gets the triangle at the specified index.

.NET Syntax

Visual Basic (Declaration)

```
Function GetTriangleAtIndex( _  
    ByVal Index As System.Integer _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IDisplayData  
Dim Index As System.Integer  
Dim value As System.Object  
  
value = instance.GetTriangleAtIndex(Index)
```

C#

```
System.object GetTriangleAtIndex(  
    System.int Index  
)
```

C++/CLI

```
System.Object^ GetTriangleAtIndex(  
& System.int Index  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Index
Zero-based index of the desired triangle

Return Value

Array of doubles (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [DisplayData::GetTriangleAtIndex](#).

Remarks

The return value is the following array of doubles:

[*vertexPt1*[3], *vertexPt2*[3], *vertexPt3*[3], *isFilled*, *lineType*]

where:

<i>vertexPt1</i> [3]	First XYZ vertex point
<i>vertexPt2</i> [3]	Second XYZ vertex point
<i>vertexPt3</i> [3]	Third XYZ vertex point
<i>isFilled</i>	Boolean returned as a double that is True if the triangle is filled, false otherwise
<i>lineType</i>	Line type as defined in swLineTypes_e

See Also

[IDisplayData Interface](#)
[IDisplayData Members](#)
[IDisplayData::IGetTriangleAtIndex Method](#)
[IDisplayData::GetTriangleCount Method](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

GetTriangleCount Method (IDisplayData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IDisplayData Interface](#) : GetTriangleCount Method (IDisplayData)

Gets the number of triangles.

.NET Syntax

Visual Basic (Declaration)

```
Function GetTriangleCount() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IDisplayData  
Dim value As System.Integer
```

```
value = instance.GetTriangleCount()
```

C#

```
System.int GetTriangleCount()
```

C++/CLI

```
System.int GetTriangleCount();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Number of triangles

Visual Basic for Applications (VBA) Syntax

See [DisplayData::GetTriangleCount](#).

See Also

[IDisplayData Interface](#)
[IDisplayData Members](#)
[IDisplayData::GetTriangleAtIndex Method](#)
[IDisplayData::IGetTriangleAtIndex Method](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

IGetArcAtIndex Method (IDisplayData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IDisplayData Interface](#) : IGetArcAtIndex Method (IDisplayData)

Obsolete. Superseded by [IDisplayData::IGetArcAtIndex2](#).

Visual Basic Syntax

Visual Basic (Declaration)

```
Function IGetArcAtIndex(  
    ByVal Index As System.Integer  
) As System.Double
```

Visual Basic (Usage)



```
Dim instance As IDisplayData  
Dim Index As System.Integer  
Dim value As System.Double  
  
value = instance.IGetArcAtIndex(Index)
```

C#

```
System.double IGetArcAtIndex(  
    System.int Index  
)
```

C++/CLI

```
System.double IGetArcAtIndex(  
& System.int Index  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Index

Visual Basic for Applications (VBA) Syntax

See [DisplayData::IGetArcAtIndex](#).

See Also

[IDisplayData Interface](#)
[IDisplayData Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

IGetArcAtIndex2 Method (IDisplayData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IDisplayData Interface](#) : IGetArcAtIndex2 Method (IDisplayData)

Gets information for the specified arc.

Visual Basic Syntax

Visual Basic (Declaration)

```
Function IGetArcAtIndex2(  
    ByVal Index As System.Integer  
) As System.Double
```

Visual Basic (Usage)



```
Dim instance As IDisplayData  
Dim Index As System.Integer  
Dim value As System.Double  
  
value = instance.IGetArcAtIndex2(Index)
```

C#

```
System.double IGetArcAtIndex2(  
    System.int Index  
)
```

C++/CLI

```
System.double IGetArcAtIndex2(  
& System.int Index  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Index

Index of the desired arc where the index begins at zero

Return Value

- in-process, unmanaged C++: Pointer to an array of doubles (see **Remarks**)
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

Remarks

The return value is the following array of doubles :

[*color*, *lineType*, *Unused*, *Unused*, *startPt[3]*, *endPt[3]*, *centerPt[3]*, *arcNormal[3]*, *rotationDir*]

where:

<i>color</i>	COLORREF returned as an integer; return value can be 0 or -1 for default color
<i>LineType</i>	Line type as defined in swLineTypes_e
<i>startPt[3]</i>	XYZ arc start point
<i>endPt[3]</i>	XYZ arc end point
<i>centerPt[3]</i>	XYZ arc center point
<i>arcNormal[3]</i>	XYZ arc normal vector
<i>rotationDir</i>	Boolean returned as a double that represents the rotation direction, where CCW = True and CW = false

See Also

[IDisplayData Interface](#)
[IDisplayData Members](#)
[IDisplayData::GetArcAtIndex2 Method](#)
[IDisplayData::GetArcCount Method](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

IGetArrowHeadAtIndex Method (IDisplayData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IDisplayData Interface](#) : IGetArrowHeadAtIndex Method (IDisplayData)

Gets information on the specified arrow head.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetArrowHeadAtIndex( _  
    ByVal Index As System.Integer _  
) As System.Double
```

Visual Basic (Usage)

```
Dim instance As IDisplayData  
Dim Index As System.Integer
```



```
Dim value As System.Double

value = instance.IGetArrowHeadAtIndex(Index)
```

C#

```
System.double IGetArrowHeadAtIndex(
    System.int Index
)
```

C++/CLI

```
System.double IGetArrowHeadAtIndex(
    & System.int Index
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Index
Index of the desired arrow head where the index begins at zero

Return Value

- in-process, unmanaged C++: Pointer to an array of doubles (see **Remarks**)
- VBA, VB.NET, C#, and C++/CLI: Not supported
See [In-process Methods](#) for details about this type of method.

Remarks

The return value is the following array of doubles:

[*arrowHeadPt[3]*, *arrowHeadDir[3]*, *arrowHeadWidth*, *arrowHeadHeight*, *arrowHeadStyle*]

where:

<i>arrowHeadPt[3]</i>	XYZ arrow head tip location
<i>arrowHeadDir[3]</i>	XYZ arrow head direction
<i>arrowHeadWidth</i>	Arrow head width measured along the arrow head direction
<i>arrowHeadHeight</i>	Arrow head height
<i>arrowHeadStyle</i>	Arrow head style as defined in swArrowStyle_e

See Also

[IDisplayData Interface](#)
[IDisplayData Members](#)
[IDisplayData::GetArrowHeadAtIndex Method](#)
[IDisplayData::GetArrowHeadCount Method](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetEllipseAtIndex Method (IDisplayData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IDisplayData Interface](#) : IGetEllipseAtIndex Method (IDisplayData)

Obsolete. Superseded by [IDisplayData::IGetEllipseAtIndex2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IGetEllipseAtIndex( _
    ByVal Index As System.Integer _
) As System.Double
```

Visual Basic (Usage)



```
Dim instance As IDisplayData
Dim Index As System.Integer
Dim value As System.Double

value = instance.IGetEllipseAtIndex(Index)
```

C#

```
System.double IGetEllipseAtIndex(
    System.int Index
)
```

C++/CLI

```
System.double IGetEllipseAtIndex(
    & System.int Index
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Index

Visual Basic for Applications (VBA) Syntax

See [DisplayData::IGetEllipseAtIndex](#).

See Also

[IDisplayData Interface](#)
[IDisplayData Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

IGetEllipseAtIndex2 Method (IDisplayData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IDisplayData Interface](#) : IGetEllipseAtIndex2 Method (IDisplayData)

Gets information for the specified ellipse.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetEllipseAtIndex2( _
    ByVal Index As System.Integer _
) As System.Double
```

Visual Basic (Usage)



```
Dim instance As IDisplayData
Dim Index As System.Integer
Dim value As System.Double

value = instance.IGetEllipseAtIndex2(Index)
```

C#

```
System.double IGetEllipseAtIndex2(
    System.int Index
)
```

C++/CLI

```
System.double IGetEllipseAtIndex2(
    & System.int Index
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Index

Index of the desired ellipse where the index begins at zero

Return Value

- in-process, unmanaged C++: Pointer to an array of doubles (see **Remarks**)
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

 Remarks

The return value is the following array of doubles:

[*color*, *lineType*, *Unused*, *Unused*, *StartPt*[3], *EndPt*[3], *CenterPt*[3], *MajorPt*[3], *MinorPt*[3],]

where:

<i>color</i>	COLORREF returned as an integer. Return value could be 0 or -1 for default color.
<i>LineType</i>	Line type as defined in swLineTypes_e .
<i>StartPt</i> [3]	Array of 3 doubles (X,Y,Z) describing the ellipse start point.
<i>EndPt</i> [3]	Array of 3 doubles (X,Y,Z) describing the ellipse end point. If the ellipse is closed, then this will be the same point as the StartPt.
<i>CenterPt</i> [3]	Array of 3 doubles (X,Y,Z) describing the ellipse center point.
<i>MajorPt</i> [3]	Array of 3 doubles (X,Y,Z) describing a point on the ellipse and the major axis.
<i>MinorPt</i> [3]	Array of 3 doubles (X,Y,Z) describing a point on the ellipse and the minor axis.

 See Also

[IDisplayData Interface](#)
[IDisplayData Members](#)
[IDisplayData::GetEllipseAtIndex2 Method](#)
[IDisplayData::GetEllipseCount Method](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetLineAtIndex Method (IDisplayData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IDisplayData Interface](#) : IGetLineAtIndex Method (IDisplayData)

Obsolete. Superseded by [IDisplayData::IGetLineAtIndex2](#).

 .NET Syntax

Visual Basic (Declaration)

```
Function IGetLineAtIndex( _  
    ByVal Index As System.Integer _  
) As System.Double
```

Visual Basic (Usage)



```
Dim instance As IDisplayData  
Dim Index As System.Integer  
Dim value As System.Double  
  
value = instance.IGetLineAtIndex(Index)
```

C#

```
System.double IGetLineAtIndex(  
    System.int Index  
)
```

C++/CLI

```
System.double IGetLineAtIndex(  
& System.int Index  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Index

Visual Basic for Applications (VBA) Syntax

See [DisplayData::IGetLineAtIndex](#).

See Also

[IDisplayData Interface](#)
[IDisplayData Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

IGetLineAtIndex2 Method (IDisplayData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IDisplayData Interface](#) : IGetLineAtIndex2 Method (IDisplayData)

Obsolete. Superseded by [IDisplayData::IGetLineAtIndex3](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IGetLineAtIndex2( _  
    ByVal Index As System.Integer _  
) As System.Double
```

Visual Basic (Usage)



```
Dim instance As IDisplayData  
Dim Index As System.Integer  
Dim value As System.Double
```

```
value = instance.IGetLineAtIndex2(Index)
```

C#

```
System.double IGetLineAtIndex2(  
    System.int Index  
)
```

C++/CLI

```
System.double IGetLineAtIndex2(  
& System.int Index  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Index

Index of the desired line where the index begins at zero

Return Value

Pointer to an array of doubles (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [DisplayData::IGetLineAtIndex2](#).

Remarks

The return value is the following array of doubles:

[*color*, *lineType*, *Unused*, *Unused*, *startPt*[3], *endPt*[3]]

where:

<i>color</i>	COLORREF returned as an integer; return value can be 0 or -1 for default color
<i>lineType</i>	Line type as defined in swLineTypes_e
<i>startPt</i> [3]	x, y, z line start point
<i>endPt</i> [3]	x, y, z line end point

See Also

[IDisplayData Interface](#)
[IDisplayData Members](#)
[IDisplayData::GetLineCount Method](#)
[IDisplayData::GetLineAtIndex2 Method](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetLineAtIndex3 Method (IDisplayData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IDisplayData Interface](#) : IGetLineAtIndex3 Method (IDisplayData)

Gets information for the specified line.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetLineAtIndex3(  
    ByVal Index As System.Integer _  
) As System.Double
```

Visual Basic (Usage)



```
Dim instance As IDisplayData  
Dim Index As System.Integer  
Dim value As System.Double  
  
value = instance.IGetLineAtIndex3(Index)
```

C#

```
System.double IGetLineAtIndex3(  
    System.int Index  
)
```

C++/CLI

```
System.double IGetLineAtIndex3(  
&    System.int Index  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Index

Index of the desired line where the index begins at 0

Return Value

- in-process, unmanaged C++: Pointer to an array of doubles (see **Remarks**)
- VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

Remarks

The return value is the following array of doubles:

[*color*, *lineType*, *lineStyle*, *lineWeight*, *startPt*[3], *endPt*[3]]

where:

<i>color</i>	COLORREF returned as an integer; return value can be 0 or -1 for default color
<i>lineType</i>	Line type as defined in swLineTypes_e
<i>lineStyle</i>	Line style as defined in swLineStyles_e
<i>lineWeight</i>	Line thickness as defined in swLineWeights_e
<i>startPt</i> [3]	x, y, z line start point
<i>endPt</i> [3]	x, y, z line end point

See Also

[IDisplayData Interface](#)
[IDisplayData Members](#)
[IDisplayData::GetLineAtIndex3 Method](#)
[IDisplayData::GetLineCount Method](#)

Availability

SOLIDWORKS 2008 SP3, Revision Number 16.3

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetParabolaAtIndex Method (IDisplayData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IDisplayData Interface](#) : IGetParabolaAtIndex Method (IDisplayData)

Gets the information for the specified parabola.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetParabolaAtIndex( _  
    ByVal Index As System.Integer _  
) As System.Double
```

Visual Basic (Usage)



```
Dim instance As IDisplayData  
Dim Index As System.Integer  
Dim value As System.Double  
  
value = instance.IGetParabolaAtIndex(Index)
```

C#

```
System.double IGetParabolaAtIndex(  
    System.int Index  
)
```

C++/CLI

```
System.double IGetParabolaAtIndex(  
    & System.int Index  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Index
[GetPolygonCount Method](#) Index of parabola

Return Value

- in-process, unmanaged C++: Pointer to an array of doubles (see **Remarks**)
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

Remarks

The return values are in an array of 16 doubles:
[*Color*, *LineType*, *LineStyleIndex*, *LineWeight*, *StartPt[3]*, *EndPt[3]*, *FocusPt[3]*, *ApexPt[3]*]
where:

<i>Color</i>	COLORREF returned as an integer; return value can be 0 or -1 for the default color
<i>LineType</i>	Line type as defined in swLineTypes_e
<i>LineStyleIndex</i>	Index location of this line style inside SOLIDWORKS Line Style Manager
<i>LineWeight</i>	Integer value defining the line weight
<i>StartPt[3]</i>	Array of 3 doubles (X,Y,Z) describing the parabola start point
<i>EndPt[3]</i>	Array of 3 doubles (X,Y,Z) describing the parabola end point
<i>FocusPt[3]</i>	Array of 3 doubles (X,Y,Z) describing the parabola focus point
<i>ApexPt[3]</i>	Array of 3 doubles (X,Y,Z) describing the parabola apex point

This set of data repeats for each parabola in the view. The size of the array is (NumParabolas * 16). Use [IDisplayData::GetParabolaCount](#) to determine the number of parabolas.

See Also

- [IDisplayData Interface](#)
- [IDisplayData Members](#)
- [IDisplayData::GetParabolaAtIndex Method](#)
- [IDisplayData::GetParabolaCount Method](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetPointAtIndex Method (IDisplayData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IDisplayData Interface](#) : IGetPointAtIndex Method (IDisplayData)

Gets information about the specified point.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetPointAtIndex( _  
    ByVal Index As System.Integer _  
) As System.Double
```

Visual Basic (Usage)



```
Dim instance As IDisplayData  
Dim Index As System.Integer  
Dim value As System.Double  
  
value = instance.IGetPointAtIndex(Index)
```

C#

```
System.double IGetPointAtIndex(  
    System.int Index  
)
```

C++/CLI

```
System.double IGetPointAtIndex(  
& System.int Index  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Index
Index of the point

Return Value

- in-process, unmanaged C++: Pointer to an array of doubles (see **Remarks**)
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

Visual Basic for Applications (VBA) Syntax

See [DisplayData::IGetPointAtIndex](#).

Remarks

The return values are in an array of 7 doubles:

[*Color*, *LineType*, *LineStyleIndex*, *PointStyle*, *CircleFilled*, *X*, *Y*, *Z*]

where:

<i>Color</i>	COLORREF returned as an integer; return value can be 0 or -1 for the default color
<i>LineType</i>	Line type as defined in swLineTypes_e
<i>PointStyle</i>	Point style as defined in swPointStyle_e
<i>CircleFilled</i>	0 if the circle is filled, 1 if not
<i>X</i>	X coordinate of the specified point
<i>Y</i>	Y coordinate of the specified point
<i>Z</i>	Z coordiante of the specified point

Use [IDisplayData::GetPointCount](#) to determine the number of points.

See Also

[IDisplayData Interface](#)
[IDisplayData Members](#)
[IDisplayData::GetPointAtIndex Method](#)

Availability

SOLIDWORKS 2009 FCS, Revision Number 17.0

SOLIDWORKS API Help

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IGetPolygonAtIndex Method (IDisplayData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IDisplayData Interface](#) : IGetPolygonAtIndex Method (IDisplayData)

Gets information for the specified polygon.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetPolygonAtIndex( _  
    ByVal Index As System.Integer _  
) As System.Double
```

Visual Basic (Usage)



```
Dim instance As IDisplayData  
Dim Index As System.Integer  
Dim value As System.Double  
  
value = instance.IGetPolygonAtIndex(Index)
```

C#

```
System.double IGetPolygonAtIndex(  
    System.int Index  
)
```

C++/CLI

```
System.double IGetPolygonAtIndex(  
& System.int Index  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Index

Index of the desired polygon where the index begins at zero

Return Value

- in-process, unmanaged C++: Pointer to an array of doubles (see **Remarks**)
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

Remarks

Format of return values is an array of doubles with the format:

[*Color*, *LineType*, *Unused*, *Unused*, *NumPolyPoints*, [x,y,z]]

where:

<i>Color</i>	Polygon color
<i>LineType</i>	Line type as defined in swLineTypes.e . You can use this index value with IDrawingDoc::GetLineFontInfo2 and IDrawingDoc::GetLineFontName2
<i>NumPolyPoints</i>	Number of XYZ points found in the [x,y,z] return value
[x,y,z]	Array of NumPolyPoints points used to describe the polygon

Use [IDisplayData::GetPolygonSizeAtIndex](#) to determine the number of elements in this array.

See Also

[IDisplayData Interface](#)
[IDisplayData Members](#)
[IDisplayData::GetPolygonAtIndex Method](#)
[IDisplayData::GetPolygonCount Method](#)
[IDisplayData::GetPolygonSizeAtIndex Method](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetPolylineAtIndex Method (IDisplayData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IDisplayData Interface](#) : [IGetPolylineAtIndex Method \(IDisplayData\)](#)

Obsolete. Superseded by [IDisplayData::IGetPolylineAtIndex2](#).

Visual Basic Syntax

Visual Basic (Declaration)

```
Function IGetPolylineAtIndex( _  
    ByVal Index As System.Integer _  
) As System.Double
```

Visual Basic (Usage)



```
Dim instance As IDisplayData  
Dim Index As System.Integer  
Dim value As System.Double  
  
value = instance.IGetPolylineAtIndex(Index)
```

C#

```
System.double IGetPolylineAtIndex(  
    System.int Index  
)
```

C++/CLI

```
System.double IGetPolylineAtIndex(  
&    System.int Index  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Index

Visual Basic for Applications (VBA) Syntax

See [DisplayData::IGetPolylineAtIndex](#).

See Also

[IDisplayData Interface](#)
[IDisplayData Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

IGetPolylineAtIndex2 Method (IDisplayData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IDisplayData Interface](#) : IGetPolylineAtIndex2 Method (IDisplayData)

Gets information for the specified polyline.

Visual Basic Syntax

Visual Basic (Declaration)

```
Function IGetPolylineAtIndex2( _  
    ByVal Index As System.Integer _  
) As System.Double
```

Visual Basic (Usage)



```
Dim instance As IDisplayData  
Dim Index As System.Integer  
Dim value As System.Double  
  
value = instance.IGetPolylineAtIndex2(Index)
```

C#

```
System.double IGetPolylineAtIndex2(  
    System.int Index  
)
```

C++/CLI

```
System.double IGetPolylineAtIndex2(  
& System.int Index  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Index

Index of the desired polyline where the index begins at zero

Return Value

- in-process, unmanaged C++: Pointer to an array of doubles (see **Remarks**)
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

Remarks

Format of return values is an array of doubles with the format:

[*polyLineType*, *DataSize*, *Color*, *LineFont*, *Unused*, *Unused*, *NumPolyPoints*, [x,y,z]]

where:

<i>polyLineType</i>	Underlying geometry type: • 0 - Polyline type • 1 - Arc or Circle type
<i>DataSize</i>	Number of elements in the GeomData array. For Type = 0, this argument is 0.
<i>Color</i>	Polyline color.
<i>LineFont</i>	Polyline font information. You can use his value with IDrawingDoc::GetLineFontInfo2 and IDrawingDoc::GetLineFontName2 . This value is valid only if LineStyle is -1.
<i>NumPolyPoints</i>	Number of XYZ points found in the [x,y,z] return value.
[x,y,z]	Array of points used to describe the polyline. This array has <i>NumPolyPoints</i> points. This method returns an array for every polyline regardless of its type.

Use [IDisplayData::GetPolyLineSizeAtIndex2](#) to determine the number of elements returned in this array.

See Also

[IDisplayData Interface](#)
[IDisplayData Members](#)
[IDisplayData::GetPolylineAtIndex2 Method](#)
[IDisplayData::GetPolyLineCount Method](#)
[IDisplayData::GetPolygonSizeAtIndex Method](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetTextPositionAtIndex Method (IDisplayData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IDisplayData Interface](#) : IGetTextPositionAtIndex Method (IDisplayData)

Gets the text item offset relative to the note text point.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetTextPositionAtIndex( _  
    ByVal Index As System.Integer _
```

) As System.Double

Visual Basic (Usage)



```
Dim instance As IDisplayData  
Dim Index As System.Integer  
Dim value As System.Double  
  
value = instance.IGetTextPositionAtIndex(Index)
```

C#

```
System.double IGetTextPositionAtIndex(  
    System.int Index  
)
```

C++/CLI

```
System.double IGetTextPositionAtIndex(  
&    System.int Index  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Index

Index of the desired piece of text where the index begins at zero

Return Value

- in-process, unmanaged C++: Pointer to an array of doubles (see **Remarks**)
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

Remarks

The return value is the following array of doubles:

[*textPtX*, *textPtY*, *textPtZ*]

where these text position values are actually offset values from the origin of this display data.

See Also

[IDisplayData Interface](#)
[IDisplayData Members](#)
[IDisplayData::GetTextAngleAtIndex Method](#)
[IDisplayData::GetTextAtIndex Method](#)
[IDisplayData::GetTextCount Method](#)
[IDisplayData::GetTextFontAtIndex Method](#)
[IDisplayData::GetTextHeightAtIndex Method](#)
[IDisplayData::GetTextInBoxHeightAtIndex Method](#)
[IDisplayData::GetTextInBoxStyleAtIndex Method](#)
[IDisplayData::GetTextInBoxWidthAtIndex Method](#)
[IDisplayData::GetTextInvertAtIndex Method](#)
[IDisplayData::GetTextLineSpacingAtIndex Method](#)
[IDisplayData::GetTextPositionAtIndex Method](#)
[IDisplayData::GetTextRefPositionAtIndex Method](#)

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IGetTriangleAtIndex Method (IDisplayData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IDisplayData Interface](#) : IGetTriangleAtIndex Method (IDisplayData)

Gets the triangle at the specified index.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetTriangleAtIndex( _  
    ByVal Index As System.Integer _  
) As System.Double
```

Visual Basic (Usage)

```
Dim instance As IDisplayData
Dim Index As System.Integer
Dim value As System.Double

value = instance.IGetTriangleAtIndex(Index)
```

C#

```
System.double IGetTriangleAtIndex(
    System.int Index
)
```

C++/CLI

```
System.double IGetTriangleAtIndex(
    & System.int Index
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Index

Index of the desired triangle where the index begins at zero

Return Value

- in-process, unmanaged C++: Pointer to an array of doubles (see **Remarks**)
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

Remarks

The return value is the following array of doubles :

[*color*, *lineType*, *Unused*, *Unused*, *startPt[3]*, *endPt[3]*, *centerPt[3]*, *arcNormal[3]*, *rotationDir*]

where:

<i>color</i>	COLORREF returned as an integer; return value can be 0 or -1 for default color
<i>LineType</i>	Line type as defined in swLineTypes_e
<i>startPt[3]</i>	XYZ arc start point
<i>endPt[3]</i>	XYZ arc end point
<i>centerPt[3]</i>	XYZ arc center point
<i>arcNormal[3]</i>	XYZ arc normal vector
<i>rotationDir</i>	Boolean returned as a double that represents the rotation direction, where CCW = True and CW = false

See Also

[IDisplayData Interface](#)
[IDisplayData Members](#)
[IDisplayData::GetTriangleAtIndex Method](#)
[IDisplayData::GetTriangleCount Method](#)